

CROSBY, United Kingdom

WMO: 033160

Lat: 53.4972N

Lon: 3.0578W

Elev: 9

StdP: 101.22

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.3	-1.5	-5.8	2.3	-1.2	-4.4	2.6	0.0	20.8	7.7	18.9	8.0	2.4	50	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	24.6	18.4	22.4	17.5	20.7	16.7	19.2	22.9	18.2	21.2	17.4	19.9	4.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	12.9	20.4	17.0	12.2	19.6	16.2	11.6	18.9	54.6	23.0	51.5	21.2	49.0	20.0	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.8	13.1	-6.1	28.4	2.1	2.3	-7.7	30.0	-8.9	31.4	-10.1	32.7	-11.7	34.3		
(5)			-6.6	21.9	2.2	2.1	-8.1	23.4	-9.4	24.7	-10.6	25.9	-12.2	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	5.3	5.4	6.8	9.1	12.0	14.7	16.7	16.6	14.4	11.6	8.2	5.7
	DBStd	4.95	2.94	2.82	2.64	2.72	2.67	2.28	2.11	1.91	2.04	2.66	2.93	3.51
	HDD10.0	655	147	129	103	47	10	0	0	0	1	15	67	136
	HDD18.3	2857	405	362	358	277	197	111	61	61	118	209	305	393
	CDD10.0	864	1	1	3	21	73	142	208	205	133	64	12	2
	CDD18.3	22	0	0	0	0	1	3	10	7	1	0	0	0
	CDH23.3	123	0	0	0	0	8	22	58	32	3	0	0	0
	CDH26.7	19	0	0	0	0	0	2	11	6	0	0	0	0
Wind	WSAvg	6.0	6.6	6.5	5.9	5.4	5.6	5.5	5.6	5.7	5.8	6.2	6.4	6.5
Precipitation	PrecAvg	806	68	53	47	53	57	61	63	70	74	90	87	85
	PrecMax	1097	128	100	111	109	108	181	126	126	142	171	185	154
	PrecMin	595	5	12	11	15	11	21	17	18	14	28	50	31
	PrecStd	117	32	21	24	27	23	37	28	25	33	36	31	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	12.9	15.6	20.9	24.4	26.4	28.2	26.9	23.2	18.8	15.0	13.1
		MCWB	10.3	10.0	11.2	13.8	17.2	18.2	19.5	19.5	17.6	15.5	13.4	11.3
	2%	DB	10.7	11.1	13.0	17.2	21.0	22.6	24.6	23.2	20.6	17.1	13.9	12.0
		MCWB	9.3	9.4	10.2	12.2	15.6	17.1	18.6	18.3	16.5	14.6	12.5	10.4
	5%	DB	9.9	10.1	11.5	14.9	18.2	20.3	22.3	21.1	19.0	16.2	12.9	11.0
		MCWB	8.6	8.5	9.4	11.2	14.0	16.2	17.7	17.4	15.8	14.1	11.4	9.6
	10%	DB	9.1	9.3	10.4	13.3	16.3	18.6	20.5	19.7	17.9	15.3	12.0	10.0
		MCWB	8.0	7.9	8.7	10.5	12.9	15.3	16.9	16.7	15.0	13.3	10.7	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.8	12.1	14.6	18.5	19.9	20.6	20.5	18.8	16.2	13.7	11.6
		MCDB	11.6	12.1	14.4	18.6	22.8	22.6	25.7	24.6	21.8	17.9	14.6	12.7
	2%	WB	9.7	9.6	10.6	12.9	16.2	17.9	19.2	19.0	17.5	15.3	12.7	10.7
		MCDB	10.5	10.8	12.4	16.0	20.0	21.6	23.4	22.1	19.6	16.7	13.7	11.8
	5%	WB	8.9	8.8	9.8	11.7	14.7	16.6	18.2	18.1	16.4	14.5	11.7	9.7
		MCDB	9.7	9.9	11.1	14.3	17.4	19.6	21.5	20.3	18.3	15.8	12.6	10.8
	10%	WB	8.1	8.0	9.0	10.7	13.4	15.7	17.2	17.3	15.5	13.6	10.9	8.9
		MCDB	8.9	9.1	10.3	12.9	15.7	18.3	20.0	19.2	17.3	15.0	11.8	9.9
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.5	5.4	6.7	6.9	6.4	6.1	5.6	5.9	5.3	4.3	4.4
		MCDBR	4.1	4.9	7.1	9.9	11.0	10.5	10.4	8.5	8.2	5.8	4.1	4.3
		MCWBR	3.8	4.0	5.0	6.3	6.7	6.6	5.8	5.1	5.2	4.2	3.8	4.0
	5% WB	MCDBR	4.0	4.5	6.0	8.4	9.8	8.9	9.0	7.2	6.9	5.0	3.9	4.2
MCWBR		4.0	3.9	4.7	5.9	6.7	6.1	5.4	4.8	5.1	4.2	3.8	4.1	
Clear-Sky Solar Irradiance	taub	0.330	0.343	0.372	0.394	0.398	0.395	0.395	0.381	0.369	0.352	0.328	0.318	
	taud	2.314	2.373	2.306	2.265	2.301	2.340	2.345	2.399	2.432	2.436	2.379	2.288	
	Ebn at Noon	627	753	804	833	850	856	847	840	804	734	630	559	
	Edh at Noon	61	78	101	119	121	118	115	104	89	72	56	52	
All-Sky Solar Radiation	RadAvg	0.63	1.31	2.47	3.83	4.88	5.10	4.83	3.97	2.82	1.56	0.77	0.47	
	RadStd	0.06	0.14	0.19	0.34	0.36	0.46	0.47	0.25	0.21	0.13	0.08	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.84	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page